

UGED1111 Logic

Inductive Reasoning

Types of Argument

◆ A **deductive argument** (演繹論証)

incorporates the claim that it is **impossible** for the conclusion to be false given that the premises are true.

◆ An **inductive argument** (歸納論証)

incorporates the claim that it is **improbable** that the conclusion be false given that the premises are true.

Deductive Vs Inductive

Deductive:

1. 所有貓都有嘴。
2. Hello Kitty 是貓。

3. Hello Kitty 有嘴。

Inductive

1. 太陽過往每天由東邊升起。

2. 太陽明天由東方升起。

Evaluate an argument

Standard

- ❑ How the premises support the conclusion
- ❑ Whether the premises are true

- Different standards for Deductive Reasoning and Inductive Reasoning
 - Deductive: Validity and Soundness
 - Inductive :Strength and Cogency

Strength



□ Validity

- Either valid or invalid

□ Strength

- A matter of degree
- Different arguments have different degree of strength

Validity as a standard

Example 1

1. 今天太陽由東方升起。
2. 昨天太陽由東方升起。

...

-
3. 明天太陽由東方升起。

Validity as a standard

1. **95 %** 的中大學生都很聰明。

2. 小明是中大學生。

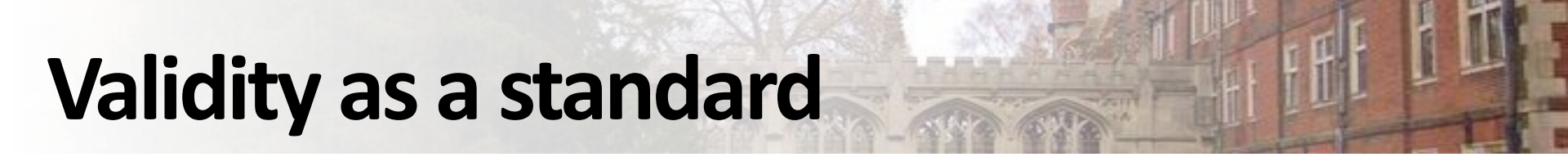
3. 小明很聰明。

1. **10 %** 的中大學生都很聰明。

2. 小明是中大學生。

3. 小明很聰明。

Validity as a standard



We do not use the concept of validity and soundness to evaluate inductive argument

-Evaluate Inductive argument

◆ Strength and cogency

Inductive Standard

□ How the premise support the conclusion

- Strength
- Strong /weak
- The argument may become weak if further evidence (premises) are added to it

□ Whether the premises are true

- Cogency
- Cogent / uncogent

Strength



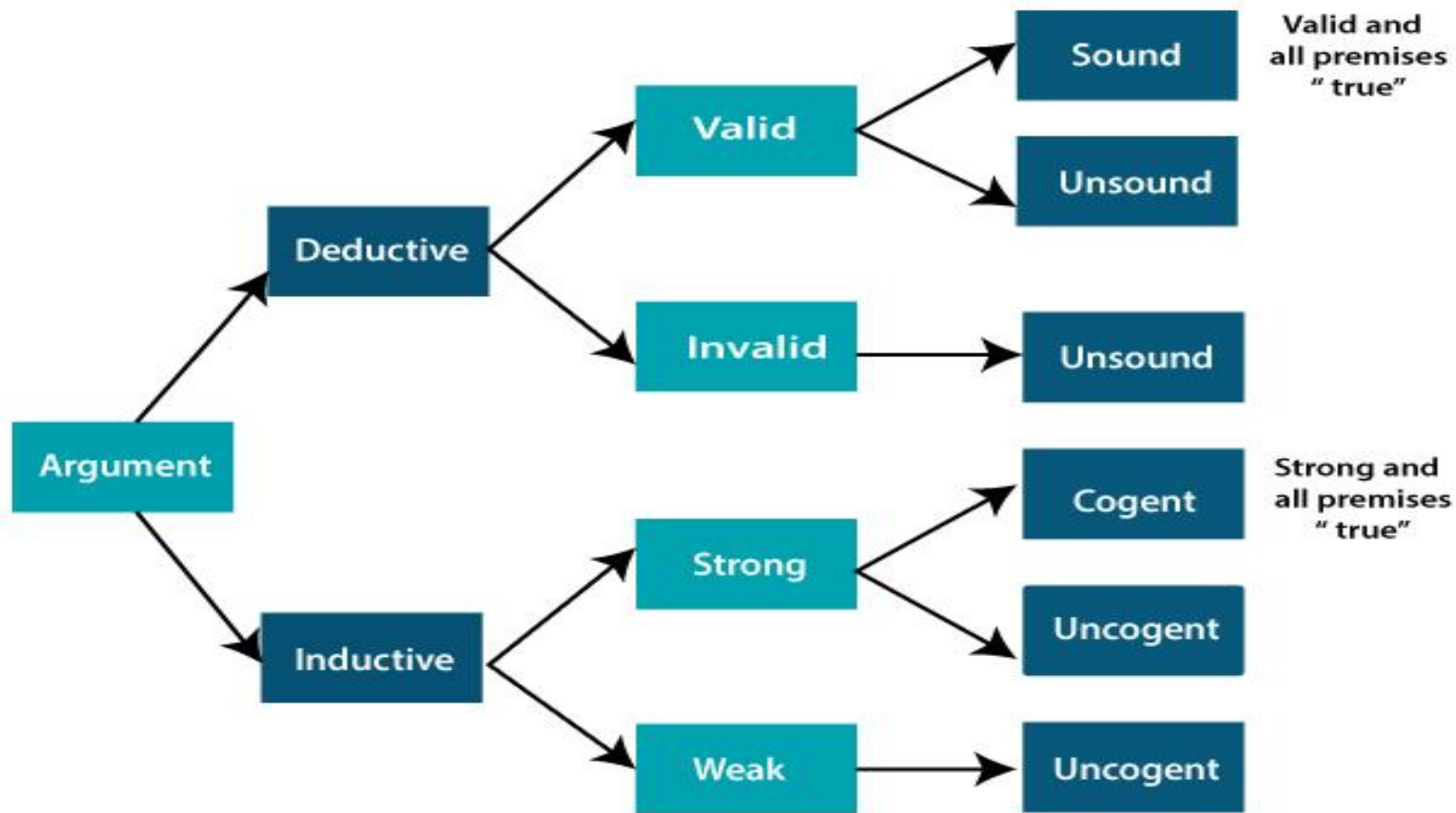
A **strong inductive argument** is an inductive argument in which it is improbable that the conclusion be false given that the premises are true. (In other words, the conclusion is more likely to be true given that the premises are true.)

A **weak inductive argument** is an argument in which the conclusion is not more probably to be true even when all premises are true.

Cogency



- ❑ Cogent and uncogent
- ❑ Strong Inductive argument + All true premises



Types of inductive reasoning

Types of inductive reasoning:

- ☐ Method of Enumerative and Statistical Syllogism**
- ☐ Argument from Authority**
- ☐ Argument from Analogy**
- ☐ Causal reasoning**

****We have different ways to evaluate the strength of different types of inductive arguments.**

Method of Enumerative

An argument using **Method of Enumerative** is to make a claims on the whole group from information provided by the sample.

1. 所有被抽查的果欄水果都沒有毒。
2. (很可能)所有果欄水果都沒有毒。

Method of Enumerative

How to evaluate an argument using Method of Enumerative

1. Is the Sample Size large enough?
2. Any Structural Bias?
 - ◆ Simple Random Sampling (簡單抽樣)
3. Is there any evidence which counter the argument?

Case

1.95%受訪牙醫表示推薦使用高露潔。

2.95%牙醫表示推薦使用高露潔

Statistical Syllogism

Statistical Syllogism is an argument which concludes claims on a particular case from a statistical statement.

1. N% of F is G

2. a is F

3. a is G

Statistical Syllogism



How to evaluate a Statistical Syllogism

1. Is N (the percentage) high enough?
2. Is there any relevant factor of that target case which is unfavorable to the conclusion?

CASE



1. 95 % 的中大學生都很聰明。
 2. 小明是中大學生。
 3. 小明很聰明。
-

Example



网上95%的名人名言
都是瞎掰的。

——鲁迅



「不要因為有配著偉人
照片，就隨便相信網路
上流傳的名言佳句。」

~ 胡適

Argument from Authority

An **argument from authority** is an argument that drawing a conclusion from the premises asserted by presumed expert.

Argument from Authority

How to evaluate an Argument from Authority?

1. Do the expert really assert/make certain claims?
2. Is the “expert” quoted a recognized expert in the field in question?
3. Is the expert’s assertion based on evidence?
4. Any other comments from other experts in the same field?
5. Is the field in question an objective research field?

Case 1

當你𩚑魚骨的時候，老人家說把碗放在頭上用筷子敲，再大喊「骨落骨落」就能解決問題，他們並表示「食鹽多過你食米」，信他們就對。所以你以後𩚑魚骨都應該用這個方法

Case

作者: amy3563867 (愛咪愛咪) 看板: Gossiping
標題: [問卦] 睡醒身上常有不知名的瘀青
時間: Tue Dec 12 01:49:09 2017

已經持續好多年了

原本分布在大腿

這兩天膝蓋下方也開始瘀青

雖然過陣子就會好

但這樣正常嗎

該裝個睡眠監視器看是不是夢遊嗎

→ ewjfd: 我家的香蕉放久也會這樣 你會不會是香蕉?
你才是香蕉 你全家都香蕉

Argument from analogy

An **argument from analogy** depends on a comparison of instances. It concludes that two instances are similar in certain aspects when the two instances have similarities in other aspects.

1. Entity A has properties a,b,c,d,m.
 2. Entity B has properties a,b,c,d.
-
3. Entity B properly has property m.

Argument from analogy

Standard for evaluation of **Argument from analogy**

1. Are they similar in relevant aspects?
2. Are there enough relevant similarities?
3. Are there any relevant dissimilarities?

Case



小明只是個三個月的嬰兒，跟他爸爸一樣是男性，頭髮不多，有手有腳。他爸爸可以喝酒，所以他也可以喝酒。

Causal Reasoning



- Arguments that concludes that A is the cause of B
- Cause and Effect
- Mill's methods (by J.S. Mill)

Causal Connection and Material Implication

- Material implication “ $A \supset B$ ”
- More requirements for A and B to be considered as **Causally Connected**
- When we says “A causes B”, it means something more than simply “ $A \supset B$ ”
- Cause is temporally prior to the effect.
- We cannot have B’s presence without A’ s presence with all other conditions fixed

Mill's Methods



- Helps us to conclude that an event/factor A is the cause of an event/factor B .
1. Method of Agreement
 2. Method of Difference
 3. Joint Method of Agreement and Difference
 4. Method of Residues
 5. Method of Concomitant Variation

Method of Agreement

- Identity the factor which is common to all occurrences with the effect's presence as the cause.
1. List out all cases of the effect occurred.
 2. Check if there is any factor which they all share.
 3. If there is a factor which is common to those cases, it is likely to be the cause.

Method of Agreement

	食物中毒	炸𩚑	蛋牛治	餐蛋麵	紅豆冰
大嘅鬼	✓	✓	✓	✓	✓
任何人	✓		✓		✓
清潔龍	✓			✓	✓
熊本熊	✓	✓	✓		✓

Which one should be the cause of食物中毒?

Method of Difference



- Identify a single factor that is present in an occurrence which the effect is present ,and absent from a case in which the effect is absent.
- confined to investigating exactly two occurrences

Method of Difference

	食物中毒	炸氹	薯條	西多	紅豆冰
大咗鬼	✓	✓	✓	✓	✓
任何人		✓	✓	✓	
清潔龍	✓	✓	✓	✓	✓
熊本熊	✓	✓	✓	✓	✓

Which one should be the cause of 食物中毒?

Joint Method of Agreement and Difference

Identify a single factor that is present in two or more cases in which the effect is present and that is absent from two or more occurrences in which the effect is absent, but never present when the effect is absent nor absent when the effect is present.

Joint Method of Agreement and Difference

	食物中毒	瑞士雞翼	芝士撈面	蒜蓉多士	凍檸茶
大嘅鬼	✓	✓	✓	✓	✓
任何人	✓	✓	✓		✓
清潔龍	✓	✓		✓	✓
熊本熊		✓		✓	

Which one should be the cause of 食物中毒?

Method of Residues



- leave the required causal connection as the “residue” by separating it from a group of known cause and effect pairs.

Method of Residues

1. 我食了血壓藥，頭痛藥和藥物C。
 2. 現在我血壓低了，頭不痛，而且沒有手震。
-
3. 藥物C很可能可以改善手震。

The method of concomitant variation

- Identifies a causal connection between two factors by matching variations in one factor with variations in another. (When factor A increases, then factor B increases, and B decreases when A decreases.)

The method of concomitant variation

- Positive correlation
 - ❑ A is positively correlated with B when B increase with A increase and B decrease with A decrease
- Negative correlation
 - ❑ A is negatively correlated with B when B increase as A decrease and B decrease as A increase
- Independence
 - ❑ B does not vary when A varies

The method of concomitant variation

- Example
- When you add more pepper into the soup, it is spicier.
- Pepper is the “cause of spiciness”

Limitation of Mill's Methods

1. True cause might not be any of the candidate causes
 - Maybe the forks are contaminated.

	食物中毒	炸氹	蛋牛治	餐蛋麵	紅豆冰
大嘅鬼	✓	✓	✓	✓	✓
任何人	✓		✓		✓
清潔龍	✓			✓	✓
熊本熊	✓	✓	✓		✓

Limitation of Mill's Methods

2. The effect might have more than one cause.

- Maybe the cause of 食物中毒 are 蛋牛治 and 餐蛋麵

	食物中毒	炸牒	蛋牛治	餐蛋麵	紅豆冰
大嘍鬼	✓	✓	✓	✓	✓
任何人	✓		✓		✓
清潔龍	✓			✓	✓
熊本熊	✓	✓	✓		✓

Limitation of Mill's Methods

3. Causation can be indeterministic.

- When 任何人腸胃太好，百毒不侵...

	食物中毒	炸肚	薯條	西多	紅豆冰
大嘅鬼	✓	✓	✓	✓	✓
任何人		✓	✓	✓	
清潔龍	✓	✓	✓	✓	✓
熊本熊	✓	✓	✓	✓	✓

Example : Sometimes people die in a heart attack but not everyone who have had a heart attack will die

Correlation and Causation

Correlation does not imply causation!

When A is positively correlated with B, there are several possibilities:

1. A causes B
2. Accidental correlation
3. B causes A
4. Both A and B are the effect of a third factor.

Reverse Causation

When A is positively correlated with B, it might not be the case that A causes B.

It might be the reverse. (B causes A)

Example:

1. Quit Smoking and death rate

2. 地方A登革熱個案上升，我們發現該地方蚊子的數量亦是上升。

Case



The sales of ice-cream is positively correlated to the number of person attacked by sharks.

Hidden common causes



When A is positively correlated with B, it might not be the case that one cause the other (A causes B or B causes A)

There might be a common cause C which cause both A and B.

Reference

- Copi, I., Cohen, C., & McMahon, K. (2011). *Introduction to Logic (14th ed. / Irving M. Copi, Carl Cohen, Kenneth McMahon ed.)*. Upper Saddle River, NJ: Pearson Education.
- Hurley, P. (2015). *A Concise Introduction to Logic (12th ed.)*. Australia ; Stamford, Ct.: Cengage Learning.
- 貝剛毅：《思方導航》（香港：滙智出版社，2011 年）

Warm-up Exercise 1

2. $\sim E$ / $\sim(E \vee \sim M)$